



DILNI WIJESINGHE

BSc (Hons) Computer Science Undergraduate

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Dilni - Portfolio

ABOUT ME

Motivated, analytical Computer Science undergraduate at IIT (University of Westminster) with foundations in algorithm design, OOP, and machine learning. Experienced in deep learning pipelines, full-stack applications, and implementing rigorous test automation and validation workflows using Java, Python, and SQL. Proven ability to clean complex datasets, evaluate model performance mathematically, and manage delivery through Agile frameworks. Seeking an Internship as an AI, Machine Learning, Software, or QA/Test Automation Engineer to contribute to high-performance development teams and drive end-to-end system reliability.

EDUCATION

BSc (Hons) Computer Science

Westminster University. (Sep 2024 – Present)

- Modules: Machine Learning, Database Systems, Algorithms, Web Application Development, Client-Server Architecture, OOP, Software Development.

GCE Advanced Level (Biological Science)

Mahanama National College Monaragala. (February 2024)

- Biology - C
- Chemistry - B
- Physics - S

TECHNICAL SKILLS

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|------------------------------|--|
| • Languages: | Python, Java, SQL |
| • AI & Machine Learning: | Supervised Learning, Deep Learning, Object Detection |
| • ML Frameworks & Libraries: | TensorFlow, OpenCV, YOLOv3, Darknet |
| • Databases & Data Formats: | MySQL |
| • Web Development: | HTML, CSS |
| • Design & Prototyping: | Figma, Canva |
| • Tools & Platforms: | GitHub, Postman, Kaggle, HackerRank |

ACHIEVEMENTS.

Collaborated as part of RobotDynamus Team to design and develop an autonomous maze-solving robot for the Micromaze 3.0 Competition (IIT RAS), implementing the Left-Hand Rule algorithm in C++ for real-time navigation, contributing to a Top 10 team placement.

CAREER INTEREST

Artificial Intelligence, Machine Learning, Data Science, Software Engineering, Business Analysis, QA / Test Automation Engineering

PROJECTS

University Projects

1. Project Coordinator & AI Developer | Harvesting Robot Arm (Harvextro Group Project)

Python, TensorFlow, OpenCV, YOLOv3-tiny, Raspberry Pi, Computer Vision

Developing an AI-powered agricultural robot that detects and harvests scotch bonnet chillies using computer vision and robotic automation. The system integrates deep learning-based object detection with a robotic arm to enable autonomous crop harvesting and sorting.

- Trained a YOLOv3-tiny object detection model for real-time chilli detection using annotated datasets.
- Implemented the image processing and detection pipeline using Python and OpenCV.
- Integrated the detection system with a Raspberry Pi-controlled robotic arm for automated harvesting.
- Collaborated within an Agile development environment, managing sprint tasks and tracking progress using ClickUp.
- Coordinated task allocation and monitored project milestones to ensure efficient team collaboration.

2. Climate Action Website - (EcoSpark - 2024)

HTML, CSS, JavaScript, Responsive Web Design

Developed a climate-action awareness website aligned with the United Nations Sustainable Development Goal (SDG 13 – Climate Action) to promote environmental responsibility and global collaboration.

- Collaborated in a team to develop an 8-page responsive climate action website.
- Built key pages using HTML, CSS, and JavaScript to present environmental data clearly.
- Worked with team members to integrate components and deliver the project successfully.

3. Data Analyst | Traffic Data Analysis Application (2024)

Python, Pandas, NumPy, Matplotlib, Data Visualization

Developed a traffic data analysis application to analyze transportation datasets and identify patterns in traffic flow.

- Collected and processed traffic datasets using Python and Pandas.
- Performed statistical analysis to identify traffic trends and patterns.
- Created data visualizations and charts using Matplotlib to present insights clearly.
- Generated reports summarizing findings to support data interpretation and decision-making.

4. Software Developer | Graph Cycle Detector Application

Java, Data Structures, Depth-First Search (DFS), Sink Elimination

Developed a standalone Java application utilizing Sink Elimination and Depth-First Search (DFS) algorithms to efficiently analyze and detect acyclicity within directed graph configurations.

- Developed a standalone Java application designed to analyze directed graphs for acyclicity.
- Implemented, analyzed, and verified two distinct algorithmic approaches: Sink Elimination and Depth-First Search (DFS).
- Conducted extensive edge-case testing with complex cyclic and acyclic graph configurations to confirm algorithmic accuracy and execution efficiency.

5. Machine Learning Engineer | Loan Approval & Limit Prediction System (2026)

Python, Pandas, NumPy, Scikit-learn, Ensemble Learning, Regression, Decision Trees

Developed an automated financial lending system to classify loan eligibility and predict maximum credit limits for approved applicants using machine learning.

- Cleaned and preprocessed raw datasets by handling missing values via mean/mode imputation, encoding categorical variables, and removing duplicate entries.
- Developed, evaluated, and optimized multiple classification models, including Logistic Regression, Gaussian Naive Bayes, and K-Nearest Neighbors (KNN).
- Engineered a Voting Classifier ensemble model that combined classification algorithms to achieve a peak loan approval prediction accuracy of 99.9%.
- Built and optimized pruned Decision Tree Regressors to forecast maximum allowed loan amounts, evaluated using MAE, RMSE, and R^2 scores.

Self Projects

6.Data Analyst | Air Quality, Weather & Respiratory Health Analysis (2024)

Python, Pandas, NumPy, Matplotlib, Data Analysis, Machine Learning

Conducted a data-driven analysis to study the relationship between environmental factors and respiratory health outcomes using real-world datasets.

- Performed data cleaning, preprocessing, and exploratory data analysis using Python and Pandas.
- Created data visualizations and correlation analysis to identify relationships between pollution, weather conditions, and hospital visits.
- Developed a linear regression model to predict respiratory-related hospital admissions.
- Generated analytical reports and visual insights to support data-driven public health decision-making.

IEEE LEADERSHIP, VOLUNTEERING & MEMBERSHIPS

Leadership & Event Management (IEEE Student Branch – IIT)

- Vice Chair – Delegate Handling Team: IEEE BotWars Competition (Ongoing)
- Standing Committee Member: Volunteer Management Committee (Since 2025)
- Delegate Handling Executive: IEEE Prestige, IX UI/UX, CodeSprint, and Sherlock 3.0 Competitions (2024 – Present)

Community Service & International Relations (Leo Club)

- Partnership & International Relations Committee Member: Leo Club of University of Ceylon Alumni (2026 – 2027)

Professional Memberships (Since 2024)

- IEEE Global Member (Since 2024)
- IEEE Societies: Robotics & Automation (RAS) | Computer Society (CS) | Communications Society (CIS))

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